In the case of reflections and translations, they produce graphs congruent to the original graph; that is, the size and the shape of the graph does not change, but in dilation, it produce graphs with shapes related to those of the original graph.

Reflection

The ***reflection*** of the graph of a function with respect to a line ℓ is the graph that is symmetric to it with respect to ℓ. A reflection is the mirror image of the graph where line ℓ is the mirror of the reflection. (See Figure 1.47.)

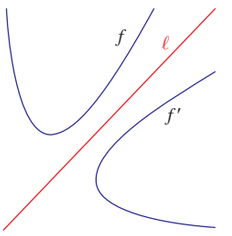


Figure 1.47

Here f' is the mirror image of f with respect to ℓ. Every point of f has a corresponding image in f'. Some useful reflections of $y = f(x)$ are

(i) The graph $y = -f(x)$ is the reflection of the graph of f about the x-axis.
(ii) The graph $y = f(-x)$ is the reflection of the graph of f about the y-axis.
(iii) The graph of $y = f^{-1}(x)$ is the reflection of the graph of f in $y = x$.

Illustration 1.5 Consider the functions:

$$(i)\ y = x^2 \quad (ii)\ y = -x^2.$$

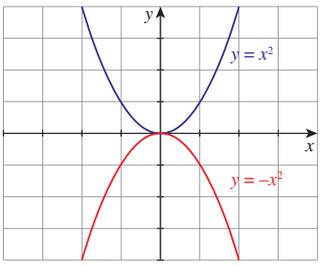


Figure 1.48

For the curve $f(x) = x^2$, $-f(x) = -x^2$. Hence, $y = -x^2$ is the reflection of $y = x^2$ about x-axis. (See Figure 1.48.)

Illustration 1.6 Consider the positive branches of

$$y^2 = x \text{ and } y^2 = -x.$$